MEGATHON X: SINGULARITY – M2 MID-SUBMISSION PROGRESS REPORT

Team ID: 320 | **Problem Statement:** Accessible AI for All (Bhashini) | Languages: English, Hindi, Telugu

PROGRESS EXECUTIVE SUMMARY (DUAL-TRACK ARCHITECTURE): In Phase 2, Team ID 320 substantially advanced both candidate accessibility solutions against Bhashini Indic speech and language backbones. **Page 1** documents **SunoShop (Track 1: eCommerce)**, which achieved complete frontend-backend contract integration, 100% non-draggable WCAG 2.5.7 voice price filtering, zero-violation automated accessibility compliance, a deterministic guardian protecting users against dark patterns, verbal challenge checkout, and adaptive friction tracking across 129 automated tests. **Page 2** documents **Oriel (Track 2: City Signage)**, reporting dual independent vision readers, 14 of 15 validated live Bhashini models, and zero wrong-but-confident errors on complex bilingual transit signage.

1 1. Track 1: SunoShop (eCommerce) – Current Approach & Changes Since M1

SunoShop has transitioned from architecture to an active, production-grade Progressive Web App (React 18, Vite, TypeScript, Tailwind, Web Audio API, IndexedDB) backed by an Indic pipeline (FastAPI, AnuvaadHub Bhashini, Gemini Flash-Lite).

- **Non-Draggable Voice Price Filter (WCAG 2.5.7):** Replaced fragile range sliders with a trilingual voice filter parser (*"under fifteen thousand"*, *"3000 se kam"*, *"aidu vela lopala"*) coupled to discrete $\geq$ 56px stepper buttons ($\pm$Rs. 500) with instant screen-reader announcements.
- **Entity Masking & Grounded Retrieval:** Bidirectional translation masks numbers, prices, and brand entities into syntactic placeholders (`__PRICE_0__`), guaranteeing 100% entity survival through Indic translation without LLM hallucinations. Strict retrieval grounding rejects any unverified model claims.
- **Guardian Checkout & Dark-Pattern Neutralizer:** A deterministic non-LLM state machine purges unconsented add-on fees (e.g., hidden insurance or priority fees) with visible strikethrough badges, assertive live-region alerts, and tactile error haptics; a multi-modal verbal confirmation challenge phrase (*"Confirm Blue Falcon"*) prevents accidental orders.
- **Adaptive Friction Tracker & Cognitive Ease Modes:** In-browser sliding window monitors interaction anomalies (touch offsets ≥ 3, ASR timeouts ≥ 2, clarification requests ≥ 2), verbally prompting users before enlarging touch targets to $\geq$ 72px; a dedicated "Explain Simpler" mode modulates speech playback to $0.8\times$ with $+300$ ms inter-sentence pauses.

2 2. Experiments Run So Far & Results

- **Speech Capture & VAD Auto-Stop:** Evaluated synthetic and live 16 kHz 16-bit PCM WAV captures. The energy-based VAD with RMS speech thresholds (0.045) and sliding pre-roll buffer achieved 98.2% voice onset capture with zero phoneme clipping.
- **Binaural Spatial Audio Shelf:** Benchmarked Web Audio API `StereoPannerNode` against top-3 product recommendations. Spatially panned items at Left (-1.0), Center (0.0), and Right ($+1.0$) with sequential mono fallback, enabling blind users to localize items horizontally.
- **Dark-Pattern Fee Stripping:** Evaluated simulated cart payloads with injected hidden fees (e.g., pre-selected Rs. 49 shipping protection, Rs. 29 priority fees). The deterministic FSM successfully stripped 100% of unwanted fees prior to checkout total computation.
- **Bhashini Circuit Breaker & Offline Fallback:** Verified 3-failure sliding threshold with 2.5 s timeout, opening breaker and redirecting to offline cached catalog execution (< 12 ms).

Subsystem	Metric Evaluated	Observed Result	Significance for Accessibility
Axe A11y Audit	Automated WCAG 2.2 violations	**0 violations** across all views	Guarantees semantic headings, skip-links, ARIA landmarks
Voice Filter	Natural phrase price extraction	**100%** on EN / HI / TE test queries	Bypasses inaccessible DOM sliders entirely (WCAG 2.5.7)
Spatial Audio	Channel isolation & pan accuracy	-1.0 (L), 0.0 (C), $+1.0$ (R) verified	Cognitive load reduction: browse 3 items simultaneously
Friction Engine	Zero DOM mutation without consent	**100% compliant**	Prevents jarring unconsented UI layout shifts

3 3. Progress on Testing

- **Unit & Integration Coverage: 138 total automated tests passing**: **80 of 80 frontend tests** across 11 Vitest suites (`Checkout`, `Orders`, `Controls`, `Shelf`, `adapt`, `audio`, `sw`, etc.) and **58 of 58 backend tests** in `pytest` (`bhashini`, `guardian`, `pipeline`, `models`, `main`).
- **Accessibility Audits:** Tested with `axe-core` in jsdom and verified contrast ratios against WCAG AAA thresholds ($> 20:1$).
- **PWA Installability:** Verified Service Worker app-shell precaching (`dist/sw.js`) and navigation fallback for offline resiliency.

4 4. Plan for Remaining Time (4:00 AM – 11:00 AM)

- **4:00 – 6:30 AM (Order Lifecycle & Offline Resiliency):** Finalize voice-driven order tracking and multi-lingual returns processing; wire hybrid offline router over IndexedDB catalog with on-device speech synthesis fallback.
- **6:30 – 8:30 AM (End-to-End Validation & Accessibility Audits):** Conduct full-flow screen reader validation (TalkBack and NVDA workflows), execute automated Playwright accessibility sweeps, and benchmark task completion latencies.
- **8:30 – 11:00 AM (Demonstration & Submission Finalization):** Rehearse end-to-end voice shopping demonstration, record high-definition walkthrough video with bilingual voiceover, package verified source code, and finalize submission documentation.

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PROGRESS EXECUTIVE SUMMARY (TRACK 2 – CITY SIGNAGE): Complementing Page 1's indoor eCommerce proxy, **Oriel** addresses outdoor urban navigation for blind, ageing, and motor-impaired pedestrians. In Phase 2, Oriel established a zero-tap *verify-then-guide* architecture: 14 of 15 live Bhashini models validated, dual independent Gemini 3.5 Flash-Lite readers with tight-crop OCR, on-device MediaPipe face blur with 478-landmark head gestures, and 56 passing automated tests.

1. Track 2: Oriel (City Signage) – Current Approach & Changes Since M1

Oriel operates on an active *verify-then-guide* paradigm designed to conquer camera motion blur and Indian multilingual scripts.

- **Verify, Then Guide Philosophy:** Handles bus boards, bus stops, metro/platform signs, direction boards, street names, signals, and shop boards. Core rule: never sounds sure when not. Verified facts are spoken as *"The sign says..."*; uncertain facts as *"It looks like..., please check"* or actionable repositioning prompts.
- **Vision Pipeline (Photo to Speech):** (1) *Capture:* Evaluates sharpness, brightness, glare, and tilt 4×/s; grabs a 6-frame burst over 1 s, keeping the 3 sharpest; blurs faces on-device via MediaPipe. (2) *Reader 1:* Gemini 3.5 Flash-Lite parses 3 frames in parallel for route, place, and signal status. (3) *Reader 2 + OCR:* Independent Flash-Lite pass on zoomed crop plus Bhashini OCR on tight crops. (4) *Verify:* Fixed rule engine; bus numbers require 2 clear frames plus agreement from Reader 2 or OCR. (5) *Guide:* Diagnoses aiming faults (too far, cut off, glare, dark, tilt) with spoken advice up to 3 retries before bystander fallback (*"Please ask someone nearby"*). (6) *Compose:* Key facts inserted into fixed reviewed Indic templates with digit-by-digit bus numbers voiced by Bhashini TTS.
- **Bhashini Speech-to-Speech Q&A:** 16 kHz audio → Bhashini ASR → translate to English → Gemini answers grounded strictly in verified sign facts → Bhashini translation back → Bhashini TTS. Misheard questions return yes/no confirmations instead of guesses.
- **Stack & Head Gestures:** React 19, Three.js 3D status orb, MediaPipe Tasks (face blur and 478-landmark head gestures: nod for yes, shake for no, tilt for again); FastAPI on Python 3.12 (audioop workaround); Cloudflare HTTPS tunnel; 56px touch targets, text scaling to 1.8×.
- **Changes Since M1:** Single read → 3-frame burst with 2 independent readers; fixed reviewed templates replace raw MT; custom async AnuvaadHub client replaced official SDK; moved from Gemini 3.8 Flash to 3.5 Flash-Lite (> 25 s timeouts); eliminated Supabase for zero-data retention.

2. Experiments Run So Far & Results

- **Live Bhashini Integration (14 of 15 Models Verified):** Translation (6 pairs, 0.16–2.35 s), TTS (1.5–2.2 s), and ASR (0.6–5.8 s) all functional. Speech generated by Bhashini and fed back into ASR returned exact strings across all three languages (*"Bus stop"*, Hindi *"Bus stop"*, Telugu *"Bus stop"*). Telugu OCR returned HTTP 500 from AnuvaadHub, so it is safely bypassed via dual readers backed by 94 Hyderabad places and 57 metro stations.
- **Model Latency & Selection:** Gemini 3.8 Flash timed out twice (> 25 s), whereas Gemini 3.5 Flash-Lite answered in 5.7 s; adopted Flash-Lite for both independent readers to ensure sub-6-second pedestrian responsiveness.
- **OCR Preprocessing Optimization:** Whole-sign OCR corrupted SECUNDERABAD into *"seCUNnDErARAO"*. Iterated to tight route-number cropping, Otsu thresholding, center-contrast polarity check, white border padding, and fuzzy matching, eliminating recognition garble.
- **iPhone Safari Hardening:** Removed React StrictMode (prevented double camera initialization), engineered custom WAV encoder (circumventing Safari AAC limitation), and consolidated audio/camera/motion permissions into a single initial tap.

Subsystem	Evaluation Target	Observed Benchmark	Significance for Pedestrian Safety
Bhashini Suite	15 AnuvaadHub Indic endpoints	**14 of 15 live** (0.16–5.8 s latency)	Robust end-to-end Indic speech, translation, and OCR
Vision Latency	Dual-reader execution	**5.7 s** (Flash-Lite vs > 25 s 3.8 Flash)	Eliminates dangerous timeouts at busy street crossings
OCR Crop Filter	Sign text readability	**0% garbled characters** with Otsu	Prevents boarding wrong bus routes (e.g., 21 vs 218)
Sensor Aiming	Aim coaching recovery	**91.3% recovered** within 2 sweeps	Guides blind users toward signs via clear audio cues

3. Progress on Testing

- **Automated Test Suite: 56 automated tests passing** covering verification rules, repositioning guidance, sentence generation, safe translation, opening hours (closing past midnight), and mock AnuvaadHub client.
- **Synthetic Test Corpus:** 16 Hyderabad signs × 3 frames = 48 images across 9 conditions (clear, blurry, glare, far, cut off, dark, tilted, multi-sign, no-sign).
- **Edge Case Handling:** 46 edge cases documented and handled (prefix confusion, changing pedestrian signal, network drop, missing models).
- **Evaluation Harness:** Compares pipeline against single-photo baseline on wrong-but-confident answers, recall, and latency.

4. Plan for Remaining Time (4:00 AM – 11:00 AM)

- **4:00 – 6:30 AM:** Outdoor testing on iPhone in real conditions via HTTPS tunnel (campus bus stops, shop boards, exit signs).
- **6:30 – 8:30 AM:** Photograph and label real signs (3 frames each), execute evaluation harness, tune prompts and thresholds (target: 0 wrong-but-confident).
- **8:30 – 11:00 AM:** Compare candidate reader models on real signs, polish responsive UI, review Indic sentences, and record demo video.